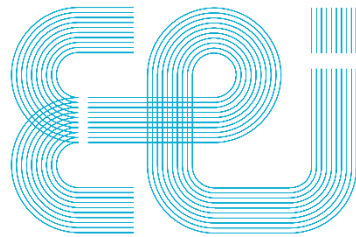


PROGRAMACIÓN TwinCAT

Lab 02

Automatización Industrial
Grado Ingeniería Industrial y Automática



El contenido de esta presentación está basado en los materiales ofrecidos por
Jakob Sagatowski en <https://www.sagatowski.com/>

ÍNDICE

- Instrucciones
- Temporizadores
- Triggers

INSTRUCCIONES

- IF/ELSE
- CASE
- FOR loops
- WHILE loops

CONDICIONAL / IF

- 1 Use **IF** to specify a block of code to be executed, if a specified condition is true
IF condition **THEN**
 // block of code to be executed if the condition is true
END_IF
- 2 Use **ELSE** to specify a block of code to be executed, if the same condition is false
IF condition **THEN**
 // block of code to be executed if the condition is true
ELSE
 // block of code to be executed if the condition is false
END_IF
- 3 Use **ELSIF** to specify a new condition to test, if the first condition is false
IF condition1 **THEN**
 // block of code to be executed if condition1 is true
ELSIF condition2 **THEN**
 // block of code to be executed if condition1 is false and condition2 is true
ELSE
 // block of code to be executed if condition1 is false and condition2 is false
END_IF
- 4 Use **CASE** to specify many alternative blocks of code to be executed

CONDICIONAL / CASE

The **CASE** statement is used to perform different actions based on a condition variable.

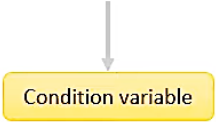
```
CASE variable OF  
  x :  
    // block of code  
  y :  
    // block of code  
  z :  
    // block of code  
ELSE  
  // block of code  
END_CASE
```

- The value of the variable is compared with the values of each case
- If there is a match, the associated block of code is executed
- If there is no match, the **ELSE** code block is executed

CONDICIONAL / CASE

The **CASE** statement is used to perform different actions based on a condition variable.

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CASE variable OF  
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ELSE  
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END_CASE
```



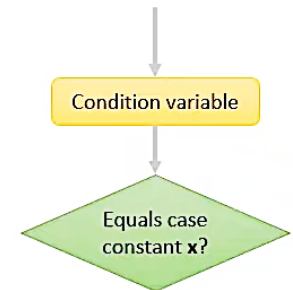
Condition variable

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    // block of code
  y :
    // block of code
  z :
    // block of code
ELSE
  // block of code
END_CASE
```

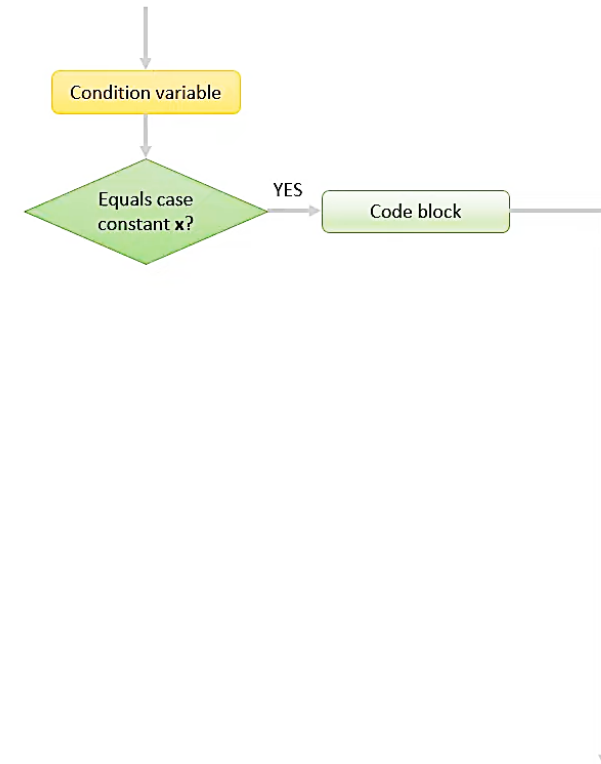


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  y : // block of code
  z : // block of code
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  // block of code
END_CASE
```



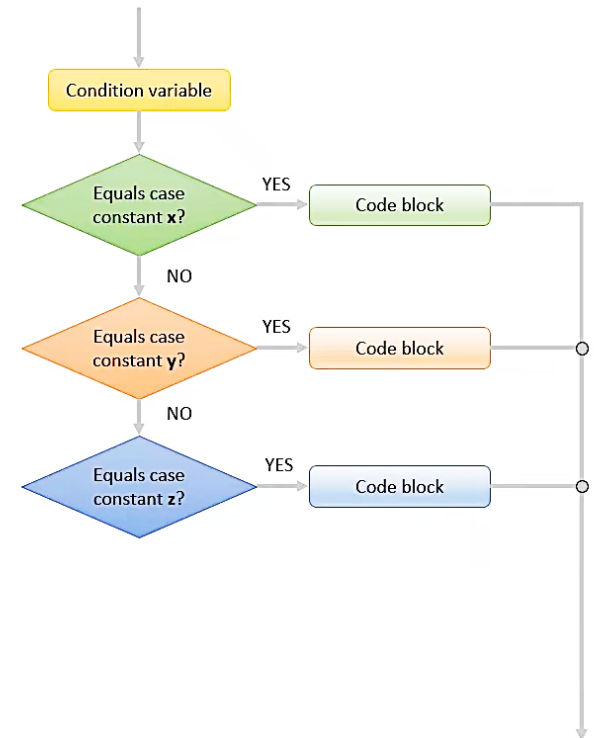
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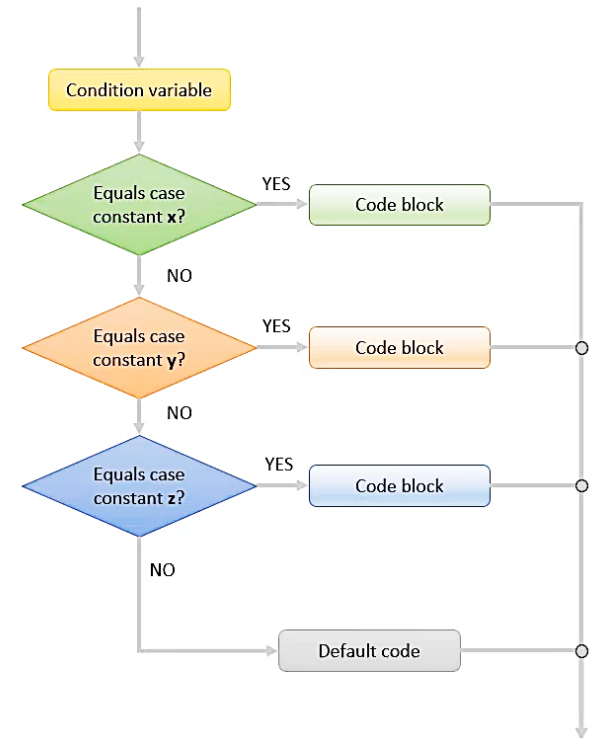


CONDICIONAL / CASE

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CASE variable OF
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  // block of code
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```

- The value of the variable is compared with the values of each case
- If there is a match, the associated block of code is executed
- If there is no match, the **ELSE** code block is executed



LOOP / FOR

FOR-loops can execute a block of code a number of times

Instead of writing:

VAR

nSum : INT;

aArray : ARRAY[1..5] OF INT;

END_VAR

nSum := nSum + aArray[1];

nSum := nSum + aArray[2];

nSum := nSum + aArray[3];

nSum := nSum + aArray[4];

nSum := nSum + aArray[5];

You can write:

VAR

nSum : INT;

nCounter : INT;

aArray : ARRAY[1..5] OF INT;

END_VAR

FOR nCounter := 1 TO 5 BY 1 DO

nSum := nSum + aArray[nCounter];

END_FOR

Optional

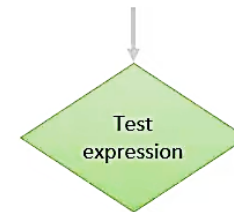
LOOP / WHILE

The **WHILE** -loop loops through a block of code as long as a specified condition is true

```
VAR
  nSum : INT;
  nCounter : INT := 1;
  aArray : ARRAY[1..5] OF INT;
END_VAR


---


WHILE nCounter < 6 DO
  nSum := nSum + aArray[nCounter];
  nCounter := nCounter + 1;
END_WHILE
```



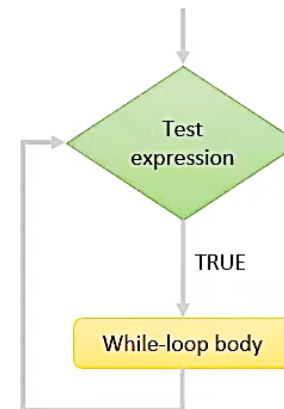
LOOP / WHILE

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  nSum : INT;
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WHILE nCounter < 6 DO
  nSum := nSum + aArray[nCounter];
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END_WHILE
```



LOOP / WHILE

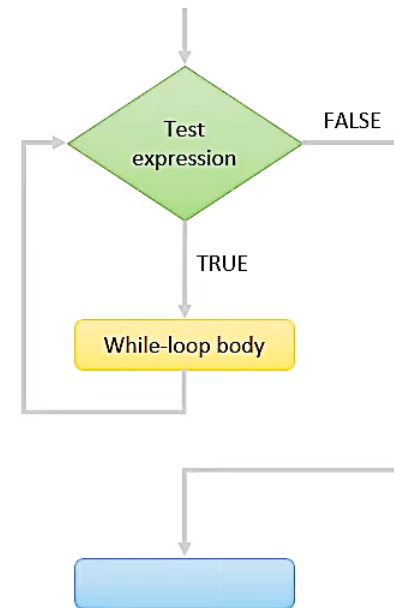
The **WHILE** -loop loops through a block of code as long as a specified condition is true

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  nSum : INT;
  nCounter : INT := 1;
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END_VAR


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WHILE nCounter < 6 DO
  nSum := nSum + aArray[nCounter];
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END_WHILE

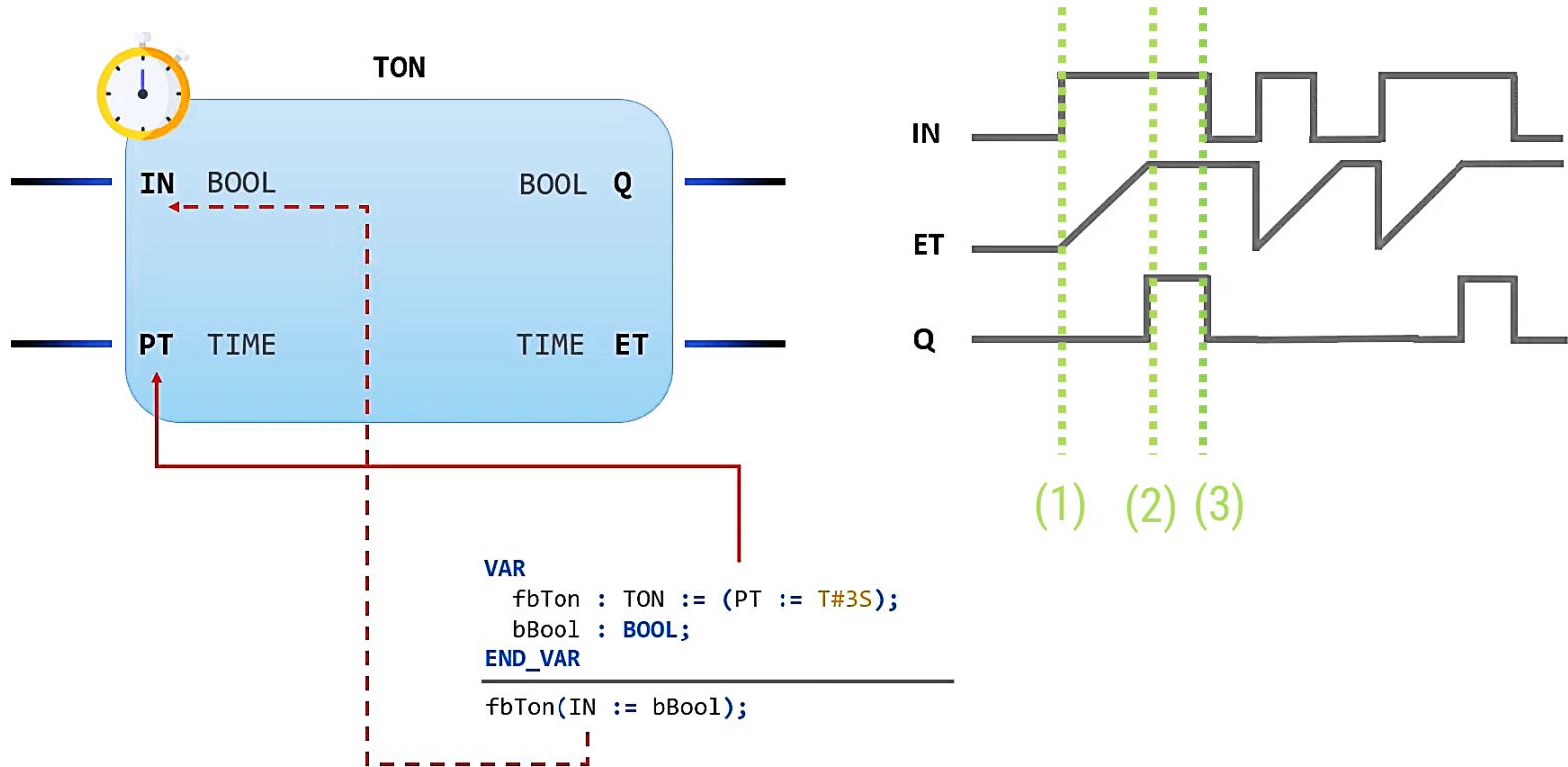
// block of code
```



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- Triggers

TON

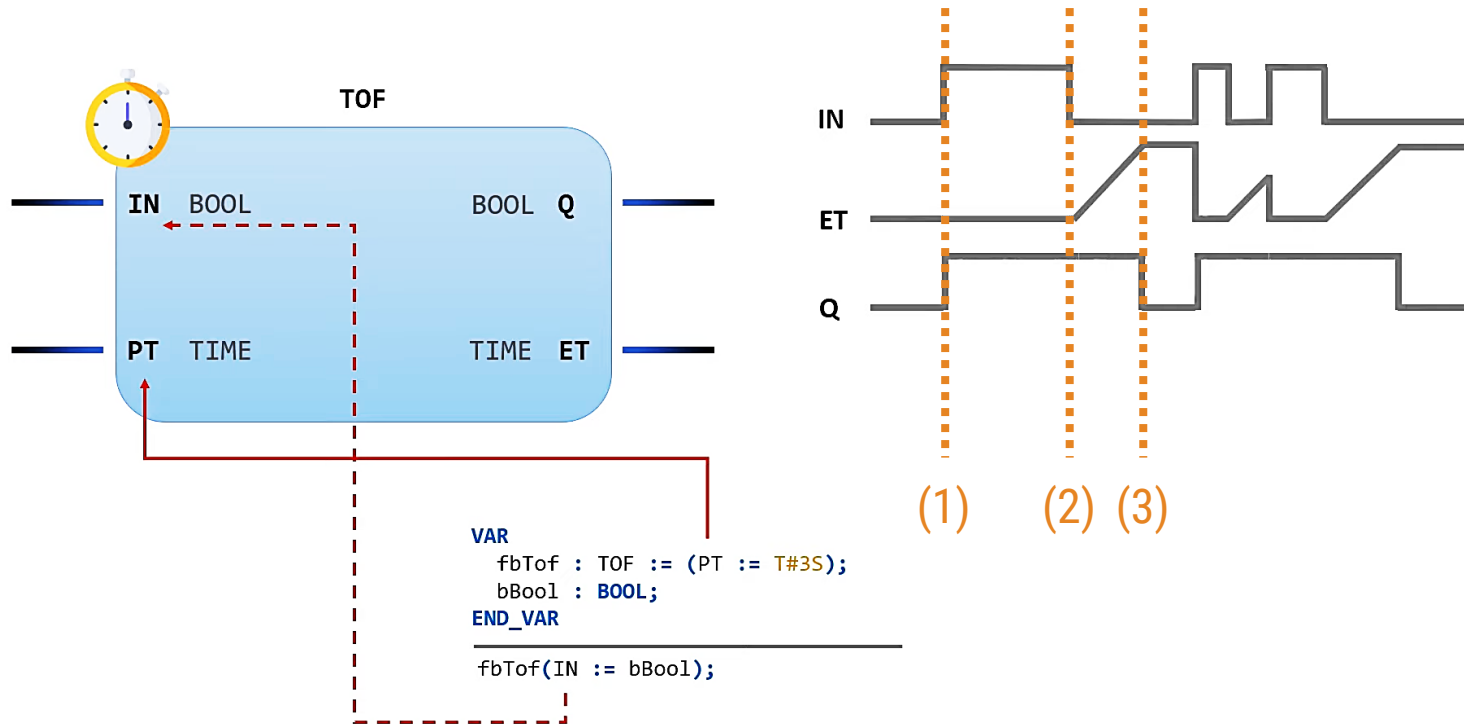


(1) ET comienza a transcurrir con flanco subida IN

(2) Una vez transcurrido $ET = PT$ se activa Q

(3) Q se desactiva **manualmente** mediante IN

TOF

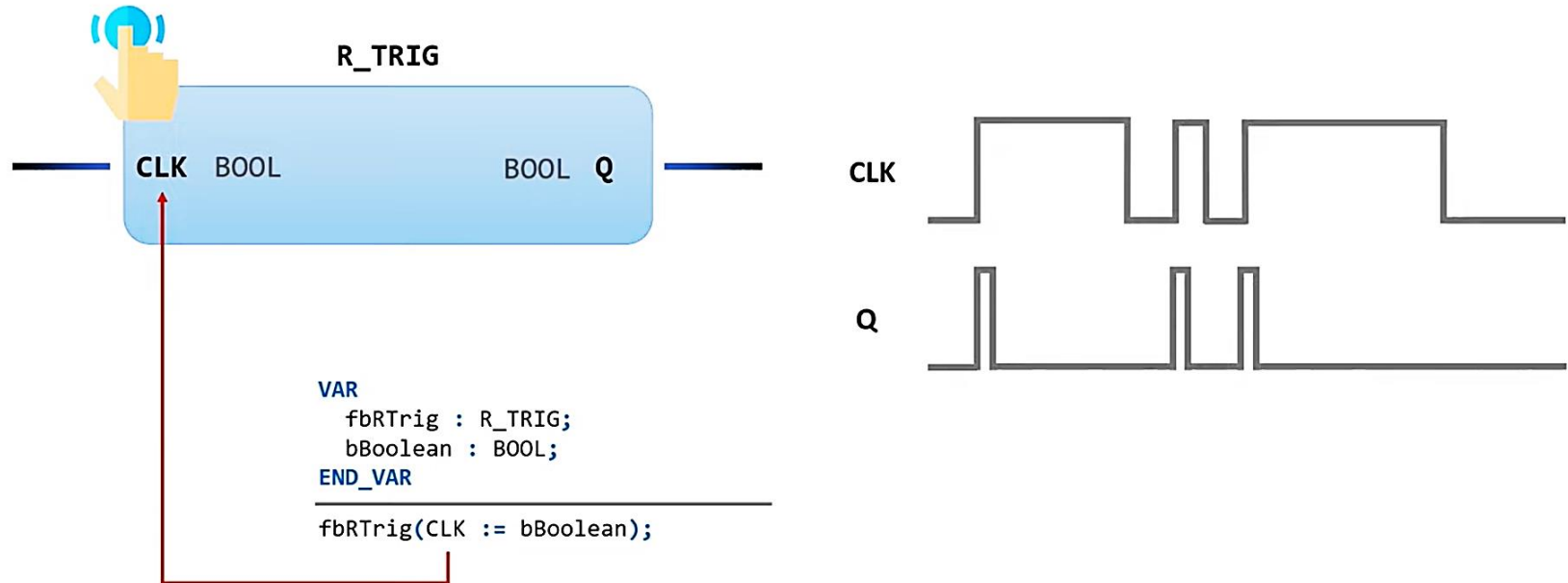


- (1) IN y Q se activan de forma simultánea
- (2) El flanco de bajada de IN activa el comienzo ET
- (3) Cuando ET=PT se desactiva Q

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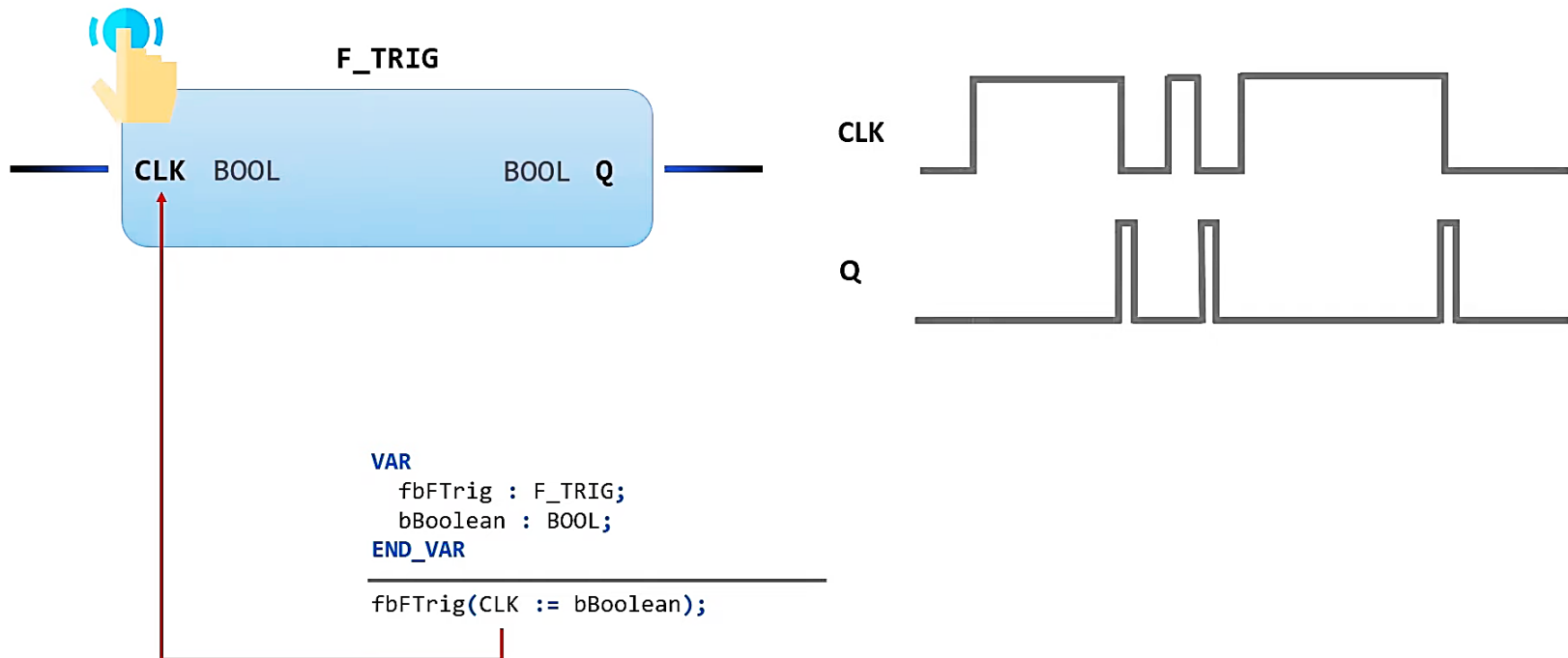
R_TRIG



Detecta flanco de subida (R = Rising) la señal CLK

Q = True, cuando se detecta el flanco de subida

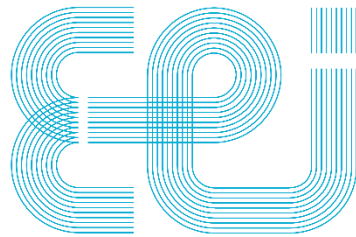
F_TRIG



Detecta flanco de bajada (F = Falling) la señal CLK

Q = True, cuando se detecta el flanco de bajada

¿PREGUNTAS?
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El contenido de esta presentación está basado en los materiales ofrecidos por
Jakob Satowski en <https://www.sagatowski.com/>